A-level

ENVIRONMENTAL SCIENCE

Paper 1 - Generated practice paper Set 10

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

• Each part opens with a line in bold saying how its marks are shared out, followed by one sentence on how to mark it.
• The figure in bold at the start of a marking point is the mark that point earns. Lines indented beneath it, starting with a dash, are the detail that belongs to the same mark; a line starting **OR** is another route to it.
• **Also accept** lists other wordings and answers that earn the mark - wording never has to match these notes, only the science. **Do not credit** lists the answers that look right and earn nothing.
• Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
• For levels-of-response questions, determine the level from the descriptors first, then position the mark within the level using the indicative content.
• Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	**Correct answer only** 1 mark for the correct option. No mark if more than one box is ticked. **1** B - Tropospheric ozone (formed by reactions of NOx in sunlight)	1
01.2	**1 mark each** 1 mark for each of the 4 points below. **1 Primary pollutant:** released directly by human activities, eg sulfur dioxide from burning coal / smelting sulfide ores – NOx from high-temperature combustion **1 Secondary pollutant:** produced by chemical reactions between primary pollutants, often with non-pollutants (water, oxygen, sunlight) **1** Sulfur dioxide is oxidised by ozone to sulfur trioxide which dissolves to form sulfuric acid – NOx dissolve to form nitric acid **1** The acids (sulfurous, sulfuric, nitric) are secondary pollutants – ozone itself is a secondary pollutant of NOx photochemistry **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round. **Do not credit** • carbon dioxide as the main cause of acid rain - the gases responsible are sulfur dioxide and oxides of nitrogen	4

Do not write outside the box

Question	Marking guidance	Marks
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 237-238; AQA spec 3.4.1.	
02.1	**2 marks per point** 2 marks for each of the 3 points below; award 1 mark where a point is only partly made. **2 High-level:** used fuel rods / fission products separated in reprocessing – vitrification in molten glass in sealed stainless steel containers, surrounded by concrete, air-cooled to remove decay heat (Sellafield) **2 Intermediate:** fuel cladding / metal tubes around fuel rods, filters from reprocessing – mixed with cement (encapsulated) in stainless steel drums, stored in concrete vaults **2 Low-level solid:** contaminated clothing, equipment – sealed in plastic bags inside steel drums / containers in a concrete-lined, monitored landfill (Drigg) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **vitrification** → sealing waste into glass **Do not credit** • 'nuclear power releases no carbon dioxide' - mining, fuel processing, construction and decommissioning all do • 'landfill gas is just carbon dioxide' - it is roughly half methane, a far stronger greenhouse gas	6
02.2	**1 mark each** 1 mark for each of the 2 points below. **1** Radioactive decay of the concentrated fission products generates heat which could damage containers **1** Isotopes with long half-lives (plutonium 24 400 years) remain hazardous for thousands of years **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
02.3	**1 mark each** 1 mark for each of the 2 points below. **1 Liquids:** filtered and passed through ion-exchange (sands / resins) to remove radionuclides, then discharged (monitored by CPA / CGM) **1 Gases:** filtered (HEPA / charcoal filters) to remove particles and iodine before release **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 303; AQA spec 3.4.3.2.14.	

Do not write outside the box

Question	Marking guidance	Marks
03.1	**1 mark each** 1 mark for each of the 3 points below. **1** Visible light passes through the atmosphere easily and is absorbed by the Earth's surface, which warms and emits infrared **1** Greenhouse gases (naturally water vapour and CO2) absorb infrared better than the average for the atmosphere, converting it to heat and re-emitting some back to the surface **1** This keeps the surface about 33 °C warmer than it would otherwise be **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **infrared** → heat radiation / long-wave radiation re-emitted by the Earth **Do not credit** • 'the greenhouse effect is bad for the planet' - the natural greenhouse effect keeps the Earth habitable; it is the enhanced effect that causes the problem	3
03.2	**1 mark each** 1 mark for each of the 3 points below. **1** CO2 is emitted in far larger quantities (every combustion process, deforestation) so its total effect is greatest **1** CO2 has a long residence time so it accumulates – methane has a short residence time (12 years) so it is removed faster **1** CFC emissions are now very small following the Montreal Protocol **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **anthropogenic** → human-caused / caused by human activity • **CFCs** → CFCs / halogenated organic compounds releasing chlorine • **infrared** → heat radiation / long-wave radiation re-emitted by the Earth **Do not credit** • answers that make ozone depletion a cause of global warming, or that describe CFCs as physically blowing a hole - chlorine atoms catalyse the breakdown of ozone and are regenerated • 'trees give us the oxygen we breathe' as the main impact - credit carbon storage, soil protection, the water cycle and habitat loss	3
03.3	**1 mark each** 1 mark for each of the 2 points below. **1** Methane has a much higher global warming potential (28x) so each tonne removed has a bigger effect **1** Its short residence time (about 12 years) means atmospheric concentrations fall quickly once emissions are cut **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **global warming potential** → warming effect compared with the same mass of carbon dioxide • **residence time** → the time a molecule stays in the atmosphere / atmospheric lifetime **Do not credit** • Points after the first 2 - the question asks for 2, so a list of extras cannot pick up the marks.	2

Do not write outside the box

Question	Marking guidance	Marks
03.4	**Any 2 of 4** 1 mark each for any 2 of the 4 points below, in any order, maximum 2. **1** Collect landfill / anaerobic digester gas and use as fuel **1** Changes in cattle diet / fewer cattle / less meat consumption **1** Draining rice paddies periodically **1** Anaerobic digestion of manure with gas capture **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks. • 'landfill gas is just carbon dioxide' - it is roughly half methane, a far stronger greenhouse gas	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 89-90; AQA spec 3.2.1.2.	
04.1	**1 mark each** 1 mark for each of the 2 points below. **1** About 40-50 traps **1** Because the running mean stops changing / fluctuating with fewer than this the mean is unreliable **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **reliable** → repeatable / gives consistent results when repeated **Do not credit** • 'repeat the readings to make them more accurate' - repeats improve reliability; accuracy depends on the method and the instrument	2
04.2	**1 mark each** 1 mark for each of the 2 points below. **1** Widely scattered results (large standard deviation) mean a single sample may not be representative **1** More samples are needed so that the effect of variability is averaged out and the mean is representative / significance can be assessed **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **standard deviation** → a measure of the spread of the data about the mean	2

Do not write outside the box

Question	Marking guidance	Marks
04.3	**Any 3 of 4** 1 mark each for any 3 of the 4 points below, in any order, maximum 3. **1** Same type and size of pitfall trap, set for the same length of time **1** Traps emptied at the same time of day / same weather conditions in both habitats **1** Same number of traps and same method of positioning (random or systematic) in each habitat **1** Other variables that could affect results (eg rainfall, temperature) monitored or controlled **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round. • **pitfall trap** → a sunken container that catches ground invertebrates • **random** → positions chosen using random numbers so the choice is not biased	3
04.4	**1 mark each** 1 mark for each of the 3 points below. **1** Flying insect activity changes with weather over minutes to weeks (temperature, wind) and diurnally **1** Samples on different occasions are needed to produce a mean that represents the typical value or to identify trends **1** The interval between samples should match the rate at which the factor changes, found from a preliminary study **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 391-393; AQA spec 3.7.1.	
05.1	**1 mark each** 1 mark for each of the 3 points below. **1** At times of low demand / surplus supply (night, windy periods) surplus electricity pumps water from the lower to the upper reservoir, storing it as gravitational potential energy **1** At peaks in demand water flows down through the turbines generating electricity **1** Can go from standby to full power in under 15 seconds so it meets sudden peaks (TV pickup) that fossil / nuclear stations cannot follow **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **HEP** → electricity generated from falling water **Do not credit** • 'nuclear power releases no carbon dioxide' - mining, fuel processing, construction and decommissioning all do • a dam described as having no environmental impacts - flooded land, blocked fish migration, trapped sediment and methane from drowned vegetation	3

Do not write outside the box

Question	Marking guidance	Marks
05.2	**Calculation** A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark. **1** Mass = $7 \times 10^6 \times 1000 = 7 \times 10^9$ kg **1** PE = m g h = $7 \times 10^9 \times 9.8 \times 500$ **1** = 3.43×10^{13} J (accept 3.4×10^{13}) **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (J) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working. • a dam described as having no environmental impacts - flooded land, blocked fish migration, trapped sediment and methane from drowned vegetation	3
05.3	**1 mark each** 1 mark for each of the 2 points below. **1** The electricity used for pumping is surplus that would otherwise be wasted / lost as heat, or cheap off-peak electricity **1** It avoids building rapid-response low-efficiency peaking plants (open-cycle gas turbines) and allows intermittent renewables to be used **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **intermittent** → only generates when the wind blows or the Sun shines / supply is not continuous **Do not credit** • 'renewables cause no pollution at all' - manufacture, construction and the land they occupy all have impacts	2
05.4	**Any 2 of 4** 1 mark each for any 2 of the 4 points below, in any order, maximum 2. **1** Two water bodies / sites with a large height difference close together (mountainous) **1** Impermeable, stable rock **1** Close to the grid / demand **1** Adequate water supply **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **Impermeable** → does not let water through **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 222; AQA spec 3.3.4.1.	

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Question	Marking guidance	Marks
06.1	**Marks shared** Up to 6 marks in total - share them between the 5 points below according to how completely each one is made. **up to 2 Habitat change:** flooding destroys the previous terrestrial / river habitats but creates a wetland which may be valuable (wetlands are uncommon) **up to 2 Wildlife barrier:** the dam blocks migrating fish (salmon, sturgeon) and prevents recolonisation along the river **up to 2 Changed river regime downstream:** regulated flow removes low-flow periods needed by species such as river turtles nesting on sandbanks and high flows that wash sediment from gravel spawning beds **up to 2 Sedimentation:** sediment settles in the reservoir instead of fertilising the floodplain and building river banks / coastlines downstream, increasing erosion **up to 2 Microclimate:** water's high heat capacity reduces temperature fluctuations – higher wind speeds – more evaporation, humidity, cloud and precipitation downwind **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • a dam described as having no environmental impacts - flooded land, blocked fish migration, trapped sediment and methane from drowned vegetation	6
06.2	**1 mark each** 1 mark for each of the 2 points below. **1** Vegetation / dead organic matter under water decays anaerobically **1** Releasing methane, a powerful greenhouse gas **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • a dam described as having no environmental impacts - flooded land, blocked fish migration, trapped sediment and methane from drowned vegetation	2
06.3	**1 mark each** 1 mark for each of the 2 points below. **1** After heavy rain surplus water is stored, reducing peak flow / flooding downstream **1** In dry periods the dam releases water to maintain river levels / supplies **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • a dam described as having no environmental impacts - flooded land, blocked fish migration, trapped sediment and methane from drowned vegetation	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 117; AQA spec 3.2.2.1.	

Do not write outside the box

Question	Marking guidance	Marks
07.1	1 mark for the point below. **1** As temperature increases the maximum dissolved oxygen concentration decreases (non-linear: falls faster at low temperatures) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1
07.2	**Calculation** 1 mark for the correct answer; working does not have to be shown. **1** About 39 °C (accept 38-40) **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (°C) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working.	1
07.3	**Calculation** A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark. **1** 9.1 - 7.9 (value at 27 °C, accept 7.8-8.0) **1** = 1.2 mg dm^{-3} (accept 1.1-1.3) **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (mg dm^{-3}) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working.	2
07.4	**1 mark each** 1 mark for each of the 2 points below. **1** Figure 1 shows the maximum / saturation values – processes that remove oxygen (aerobic decomposition of organic matter / respiration) can give values below the line **1** Higher temperature increases the metabolic rate of organisms and the rate of bacterial decomposition so oxygen is used faster – values above the line are impossible **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2

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Question	Marking guidance	Marks
07.5	**Any 3 of 6** 1 mark each for any 3 of the 6 points below, in any order, maximum 3. **1** Increased metabolic rate so more energy used / less surplus for growth **1** Eggs develop faster / hatch earlier (out of step with food supply) **1** Non-indigenous warm-water species may thrive and out-compete indigenous species **1** Toxicity of other pollutants is greater as toxins are metabolised faster **1** Reduced resistance to disease **1** Proteins / enzymes denatured if temperature is very high (above about 40 °C) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks.	3
07.6	1 mark for the point below. **1** Hot water is sprayed from a shower floor in a tall hollow tower – air absorbs the heat and rises, drawing in more air (updraught), so the water returned is cooler (and saturated with oxygen) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 256-257; AQA spec 3.4.3.2.6.	
08.1	**1 mark each** 1 mark for each of the 3 points below. **1** Metal-rich lumps 5-10 cm across on the deep seabed (4000-5000 m), containing about 30% manganese plus nickel, copper, cobalt, iron, titanium **1** Exploitation would be very expensive (remotely operated collectors, pumping to the surface) / no viable technology yet **1** International agreement on ownership of the seabed and its resources is needed **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3
08.2	**1 mark each** 1 mark for each of the 3 points below. **1** Collectors disturb the seabed and kill benthic organisms living on it **1** Separating nodules from sediment creates plumes that increase turbidity with unknown consequences **1** Re-deposition of sediment covers and kills filter feeders and other benthic organisms – slow recovery in the deep sea **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3

Do not write outside the box

Question	Marking guidance	Marks
08.3	**1 mark each** 1 mark for each of the 4 points below. **1** Used in small amounts in many essential technologies (smartphones, hybrid vehicles, wind turbines, fibre optics, LEDs) **1** Exploitable land deposits are limited and mainly in China – chemical separation is complex and produces large amounts of toxic waste **1 Bacterial adsorption:** dissolved rare earth ions adsorb onto the cell surfaces of bacteria (Bacillus subtilis) and are washed off at specific pHs to separate them **1** Deep sea sediments contain relatively high concentrations but no viable extraction method yet **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **adsorption** → nutrient ions sticking to the surfaces of clay and humus **Do not credit** • 'wind turbines can generate all the time' - output depends on wind speed, and they shut down in very high winds	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 145-146; AQA spec 3.2.3.7.	
09.1	**1 mark each** 1 mark for each of the 3 points below. **1** Shale has very low permeability so the 'tight' oil / gas cannot flow through the rock to a well **1** Fluid is pumped down at high pressure to open fissures in the shale along which oil / gas can flow to the recovery well **1** Sand grains (to hold fissures open) and solvents are pumped in to increase the recovery rate **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **hydraulic fracturing** → pumping fluid into rock at high pressure to release gas or oil • **permeability** → how easily water passes through / lets water through	3
09.2	**1 mark each** 1 mark for each of the 5 points below. **1** Natural gas may enter aquifer water **1** Injected chemicals may reach aquifers or the surface – toxic metals in the rock may be mobilised **1** Large volumes of water needed **1** Earth tremors as fracking releases natural tensions in the crust (though not tremors that could not occur naturally) **1 Reduced by:** collection, treatment and re-use of waste water – well casing standards – restricting sites away from aquifers / sensitive areas – monitoring **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **fracking** → pumping fluid into rock at high pressure to release gas or oil • **aquifer** → a permeable rock that stores groundwater / underground water store	5

Do not write outside the box

Question	Marking guidance	Marks
09.3	**1 mark each** 1 mark for each of the 2 points below. **1** Extraction is expensive and energy-intensive (quarrying and hot water treatment, steam / solvent / combustion in-situ; heating shale to release oil) so it is not economically viable at current prices **1** Reserves only include what is economically exploitable now – high environmental impacts (land take, water, CO2) also limit exploitation **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 184, 189-190; AQA spec 3.3.4.1.	

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Question	Marking guidance	Marks
10.1	**Levels of response** Decide the level from the descriptors first, then use the indicative content to place the mark within that level. The points below are not a checklist. **Level 3 (7-9 marks)**: Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. **Level 2 (4-6 marks)**: Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. **Level 1 (1-3 marks)**: Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. **0 marks**: No relevant content. **Indicative content - a guide to placing the mark, not a checklist** • **Principles:** dB is logarithmic (10 dB = 10x; 3 dB = 2x) so small reductions in dB are large reductions in intensity – effects on humans (hearing, stress, behaviour) and wildlife (breeding birds, bats, cetaceans) • **Aircraft design:** high bypass ratio engines, chevron nozzles, hush kits, acoustic liners, fairings, composites, blended wing • **Aircraft operation:** take-off angle, flight paths avoiding urban areas, constant descent, reduced reverse thrust, night restrictions, Chapter noise limits, fines and noise charges, quota count • **Airport design / location:** away from population centres, taxi and engine-test areas away from housing, acoustic walls, baffle mounds, land-use restrictions, free acoustic insulation • **Railway:** track polishing, sound-absorbing ballast and suspension, pantograph fairings, wheel lubrication, composite brakes • **Road:** porous / rubber asphalt, acoustic fences, embankments, tree planting, aerodynamics, engine insulation, traffic management and routing, double glazing • **Industrial:** silencers on compressors, pressing instead of stamping, nylon rollers, drilling instead of pile-driving, bubble curtains, timing of blasting, acoustic mats / curtains, remote operation, ear protection • **Domestic:** acoustic absorbers in appliances, volume limiters, selection of quieter appliances, ear defenders • **Evaluation:** which approaches are most effective / cost-effective – conflicts (airports near cities for convenience) – legislation and monitoring (Leq 57 dB contours) needed to enforce **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • Any accurate, relevant material that is not in the indicative content.	9
	• Any appropriate named example, including ones that are not listed here. • **quota** → a legal limit on the mass or number that may be caught **Do not credit** • quotas described as solving overfishing on their own - undersized fish are still caught and discarded	
	Model answer - Level 3 (9 marks)	

Do not write outside the box

Question	Marking guidance	Marks
	Because the decibel scale is logarithmic, a 10 dB cut is a tenfold reduction in intensity and 3 dB halves it, so modest-sounding design changes matter. Aircraft engines have been made quieter by high-bypass-ratio turbofans, chevron nozzles, hush kits and acoustic liners, with fairings and composite airframes reducing airframe noise. Operation can be changed too: steeper take-off angles and flight paths that avoid housing, constant-descent approaches, restricted reverse thrust, night curfews, Chapter noise limits enforced by fines and quota counts. Airport location and layout - away from population centres, engine-test areas and taxiways kept from housing, acoustic walls and baffle mounds, land-use planning and free insulation for residents - deal with the noise that remains.	
	Railways use track polishing, sound-absorbing ballast and suspension, pantograph fairings, wheel lubrication on curves and composite brakes. Roads use porous or rubber-crumb asphalt, acoustic fences and embankments, tree planting, engine insulation and aerodynamics, traffic routing and double glazing. Industry fits silencers to compressors, presses rather than stamps, uses nylon rollers, drills instead of pile-driving, surrounds piling with bubble curtains to protect cetaceans, times blasting, hangs acoustic mats and operates remotely, with ear protection as a last line. Domestic appliances use acoustic absorbers and volume limiters.	
	Design changes at source are the most effective because they reduce noise for everyone; operational limits depend on enforcement; location works only for new developments, and airports are still sited near cities for convenience. Legislation and monitoring of noise contours (57 dB L_{eq}) are what turn these options into actual reductions.	
	Weaker answer - Level 2 (5 marks)	
	Noise pollution can be reduced in different ways. Aircraft can be designed with quieter engines and can fly at night less often. Airports can be built away from cities. Roads can have sound barriers next to them and quieter road surfaces. Trains can have their tracks polished and use better brakes. Factories can use quieter machines and workers can wear ear defenders. Noise can also be reduced by planting trees. These methods reduce the noise that reaches people.	
	What makes the difference	
	• Level 2 gives correct but generic examples ('quieter engines'); Level 3 names the design features (high bypass ratio, chevron nozzles, acoustic liners) the question's 'named examples' requires.	
	• Level 3 organises by design / operation / location as the question asks; Level 2 mixes them.	
	• The decibel-scale point and the closing evaluation of which approaches work best lift Level 3 into the top band.	
10.2	1 mark for the point below. **1** The area within which the average (equivalent continuous) sound level between 07:00 and 23:00 in summer exceeds 57 dB - taken as the onset of significant community annoyance **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1

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Question	Marking guidance	Marks
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 286-295; AQA spec 3.4.3.2.13.	
11.1	**Evaluate the contribution of new technologies to the monitoring and understanding of environmental change in the physical environment.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • Historical data limitations – proxy data - ice cores, isotope ratios (O-18:O-16, C-14), Mg:Ca in calcite, tree rings, pollen, coral • **Satellite sensors:** visible light imagery, infrared for vegetation and temperature, radar microwaves for wind, wave height, sea level and oil, weather monitoring – passing through cloud • **Sensor carriers:** ROVs / drones, AUVs, balloons, aircraft, animals – data recorders for abiotic factors – GPS tracking of floats and sampling sites – Argo floats and ocean monitoring • Climate models and their assumptions – monitoring atmospheric composition (Mauna Loa), ozone (Dobson units), sea level and ice • Applications to hydrosphere, minerals (remote sensing surveys), soils, pollution monitoring • **Evaluation:** improved coverage and resolution vs cost, calibration, interpretation, remaining uncertainty – justified conclusion	25
	Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion. **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 9, 89-107, 402-407; AQA spec 3.2, 3.7.	

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Question	Marking guidance	Marks
11.2	**Discuss how an understanding of the hydrological cycle can be used to develop strategies for the sustainable management of water resources.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • Hydrological cycle processes and reservoirs – residence times – dynamic equilibrium – increasing demand (population, affluence, irrigation, industry) • **Human impacts on transfer rates:** deforestation, agriculture, urbanisation, climate change, abstraction – consequences for rivers, aquifers (salt incursion, subsidence, wetland loss), reservoirs • **Case studies of unsustainable exploitation:** Aral Sea, Nile, Tibetan rivers, Ogallala, North China Plain, Malta • **Sustainable strategies:** metering, low-water appliances, greywater, xeriscaping, drip irrigation, leak repair – new sources: rainwater catchment, reservoirs and estuary barrages (site factors), unexploited aquifers, inter-basin transfers, artificial recharge, river regulation, afforestation, desalination • Water treatment to make more water usable: sedimentation, flocculation, filtration, activated carbon, sterilisation, ion exchange, reverse osmosis – pollution control (sewage, AMD, buffer strips)	25
	• Links to thermohaline circulation and climate – energy costs – circular economy principles (catchment management) • Evaluation of which strategies are most sustainable in different regions – conclusion *Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion.* **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 114-128; AQA spec 3.2.2.	

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